COURSE HANDBOOK

Machine Learning

MSc Data Science / Software Engineering
UE Innovation Hub, Potsdam

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Course Overview

This course introduces core machine learning concepts, workflows, and models used in modern AI. Students will learn how to frame an ML problem, prepare data, build and evaluate models, and understand where advanced methods such as deep learning and LLMs fit in practice.

The emphasis is technical and applied. Students are expected to connect business or application needs with model choice, validation strategy, and generalization behavior.

Prerequisites	Basic programming, probability, and linear algebra.
Tools	Python and Jupyter Notebook; other libraries will be introduced during the course.
Format	Conceptual lectures, worked examples, discussion, and hands-on exercises.
References	• Deep Learning (https://www.deeplearningbook.org/) • Selected course notes, CS229 notes (https://cs229.stanford.edu/main_notes.pdf)

Course Objectives

- Provide students with a sound understanding of the end-to-end machine learning process, from problem identification and data preparation to model evaluation and deployment.
- Develop conceptual and applied knowledge of classical machine learning, ensemble learning, deep learning, and transformer-based language models.
- Enable students to diagnose model behaviour and improve performance through validation, regularization, fine-tuning, and informed model selection.

Course Learning Outcomes

- **Apply** the machine learning workflow by identifying the problem, defining data requirements, preparing datasets, splitting data appropriately, and developing baseline models.
- **Develop and evaluate** AI models using linear regression, logistic regression, neural networks, random forests, convolutional neural networks, and transformer-based language models.
- **Analyze** and improve model performance by diagnosing underfitting and overfitting, applying regularization and fine-tuning strategies, and comparing encoder, decoder, and encoder-decoder architectures.

Indicative Content Coverage

1. ML Foundations	ML process; business vs ML problem; data needs; collection; preprocessing; train/validation/test splits; evaluation.
2. Basic Models	Linear regression, logistic regression, neural networks, underfitting, overfitting, regularization, and baseline thinking.
3. Ensemble & Deep Learning	Random forest, convolutional neural networks, and practical considerations in model selection and improvement.
4. Large Language Models	Basics of NLP, transformer models, encoder/decoder/encoder-decoder architectures, LLM fine-tuning, and optional RAG and agentic AI.

Learning Approach and Expectations

- The course combines explanation, worked examples, and hands-on illustration, moving from simple baseline models to more advanced methods.
- Students should be prepared to compare models, justify design choices, and explain why a model generalizes well or poorly.
- Regular attendance and active participation are strongly encouraged. Students should plan for at least 3-4 hours of study and practice each week outside class.

Assessment Structure

Component	Weight	Notes
Quizzes	60%	Weekly or bi-weekly quizzes. No retake, in line with university policy.
Project	40%	Students must obtain at least 50% in the project component to pass the course.

Project Handbook

The course project is intended to build teamwork, structured problem understanding, and the ability to justify technical choices.

- Students will form their own teams of 2-3 members.
- A list of project topics will be shared around the 3rd or 4th teaching week.
- Each team will submit three topic preferences in ranked order.
- Topics will be assigned by the instructor; preferences will be considered, but a preferred topic cannot be guaranteed.
- The first presentation will take place in the 8th or 9th week. Each group will have 5 minutes. No implementation is required at this stage. The focus is on understanding the problem and proposing a methodology. This milestone carries 10% of the total course score.
- The final presentation will take place in the 15th and 16th week. Each group will have 10 minutes to present the technical details of the work. This milestone carries 30% of the total course score.
- Evaluation is group-based. Every student is expected to answer questions about the project, so teams should be formed carefully and free-riding is not acceptable.

Advisory note. Shared ownership of code, presentation material, and technical understanding is expected across the full team.